CUBAN

NATIONAL CHEMISTRY CONTEST 2022

"You should never be afraid of what you're doing, when it's right."

Marie Curie

Name(s): ____________________________ Last name(s): ____________________________

School: ____

Municipality: ____________________________ Province: ____________________________

CODE: ____________________________ Test: **DAY 2. ELEVENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 11^{th} grade. The exam has 2 problems (problem 7, 4 items; problem 8, 8 items) and a total of 4 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), and each item scoring (in marks) is indicated.

Success!

Ministerio de Educación

República de Cuba

FACULTAD DE QUÍMICA

UNIVERSIDAD DE LA HABANA

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro´s constant	$N_A = 6{,}0221 \cdot 10^{23}\ \mathrm{mol^{-1}}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8{,}3145\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	$273{,}15\ \mathrm{K}$	Gibbs´ free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday´s constant	$F = 96485\ \mathrm{C \cdot mol^{-1}}$	Nernst´s equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Ionic product of wáter at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$	Arrhenius´ equation	$k = A \cdot \exp(-\frac{E_A}{R \cdot T})$
Standard conditions	$p° = 100\ \mathrm{kPa}$ $c° = 1\ \mathrm{mol \cdot L^{-1}}$	$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$	

$1\ \mathrm{J} = 1\ \mathrm{C} \cdot 1\ \mathrm{V} = 1\ \mathrm{kPa} \cdot 1\ \mathrm{L} = 1\ \mathrm{W} \cdot 1\ \mathrm{s}$ $\qquad$ $1\ \mathrm{C} = 1\ \mathrm{A} \cdot 1\ \mathrm{s}$

PERIODIC TABLE OF ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 7. Hydrogen sulfide (21 points)

7.1	7.2	7.3	7.4	Total	Total
6	9	11	16	42	21

Hydrogen sulfide, H_2S, is a weak acid which can be used to precipitate metal ions from aqueous solutions by tight pH control.

7.1 Calculate the concentration of the sulfide anion in a saturated solution of hydrogen sulfide. Consider 0.1 mol of hydrogen sulfide dissolves in 1L of water at STP. The primary and secondary dissociation constants of hydrogen sulfide are $1\cdot10^{-7}$ and $1\cdot10^{-14}$ respectively.

7.2 Calculate the concentration of sulfide anion in 0.1 mol/L hydrochloric acid solution saturated with hydrogen sulfide.

7.3 If 0.1 mol/L hydrochloric acid solution containing Hg^{2+} and Bi^{3+} ions at 5 mmol/L concentration is saturated with hydrogen sulfide, which sulfide precipitates first? What is the concentration of the cation involved after precipitation?

7.4 Hg_2^{2+} ions present in the aqueous solutions of mercury(I) salts disproportionates easily. Determine by calculations if when it is saturated with hydrogen sulfide a solution of 0.1 mol/L of strong and soluble salt $Hg_2(NO_3)_2$ at pH = 1 also occurs the disproportion of the mercury(I) ions. Consider that the equation of the possible disproportion is as follows:

$$Hg_2^{2+}(aq) + H_2S\ (aq) = Hg\ (s) + HgS\ (s) + 2\ H^+(aq)$$

Data:

$K_{sp}(Bi_2S_3) = 1.6\cdot10^{-72}$ $K_{sp}(HgS) = 2.0\cdot10^{-32}$

$$Hg^{2+} \xrightarrow{\ 0.91\ V\ } Hg_2^{2+} \xrightarrow{\ 0.80\ V\ } Hg$$

$$\underbrace{Hg^{2+} \longrightarrow Hg}_{0.85\ V}$$

Problem 8. Electrolytic solutions (29 points)

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	Total	Total
4	3	7	5	2	4	2	2	29	29

Sodium hypochlorite is one of the most widely used substances for several centuries as a water disinfectant and bleach. In the process of obtaining hypochlorite, first, by means of electrolysis of a brine solution (NaCl), gaseous chlorine was produced. For this, a current of 11.5 A was passed for 2 h and 50 min to a volume of 1L thereof.

8.1 State the possible cathodic and anodic half-reactions, as well as the global equation of the process that occurs.

8.2 How much amount of substance of the chlorine substance is produced?

Chlorine that was obtained after the electrolysis was thoroughly bubbled into a 1 mol/L sodium hydroxide solution. The container where said solution was found, after adding all the chlorine, was hermetically closed.

8.3 Show by calculations that the disproportionation of chlorine in a basic medium is a spontaneous process and state the global equation of this reaction.

8.4 Calculate the pH of the final solution. Note that the limiting substance is NaOH.

From the brine solution (NaCl, 12.0% by mass) used as starting reagent, it is known that it has a density of 1.085 g/mL (at 25°C) and that when adding the salt to the water the volume of the latter increased 1.046 times.

8.5 Calculate the molar concentration of this solution.

8.6 Calculate its molality. Consider the density of water at this temperature equal to 1 g/mL.

8.7 What will be the melting point of this solution?

8.8 It was found that the solution begins to cool at -8.18°C. What is the reason for this difference between the calculated value in the previous section and the experimental one?

Data:

$K_c(H_2O) = 1.86\ K\cdot kg\cdot mol^{-1}$ $K_a(HClO) = 4.0\cdot 10^{-8}$

$$ClO_3^- \xrightarrow{0.33\ V} ClO_2^- \xrightarrow{0.67\ V} \underbrace{ClO^- \xrightarrow{?} Cl_2 \xrightarrow{1.36\ V} Cl^-}_{0.88\ V}$$